



Department of Information Technology

Academic Year 2025 – 2026 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3452 & Theory of Computation

Name of the Faculty member (s): Dr. V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit III / PDA to CFG
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO10
- **Activity Chosen:** Jigsaw
- **Justification:**
 - The conversion from Pushdown Automata (PDA) to Context-Free Grammar (CFG) involves multiple steps (states, stack symbols, transitions). The Jigsaw method breaks this complex process into smaller, manageable subtopics, making it easier for students to understand. Unlike passive lectures, Jigsaw ensures every student is actively involved, responsible for learning and explaining a portion of the topic.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - The topic “**PDA to CFG Conversion**” was taught in the class.
 - The class was divided into six teams, each consisting of ten members. The groups were formed based on students’ academic performance and abilities, and were designated as **home groups**.
 - One student from each home group was selected, and **expert groups** were formed comprising one representative from each home group.
 - Each student in the expert group was assigned a specific subtopic. A duration of 10 minutes was allotted for discussion, during which the course instructor provided clear instructions to guide the activity.
 - After the expert group discussions, the groups were dissolved, and all students returned to their respective home groups to continue collaborative learning.
 - Within the home groups, each member acted as an expert in a particular aspect of PDA to CFG conversion and shared their knowledge with the group. This ensured that all members gained a comprehensive understanding of the topic.
 - This approach enabled students to develop in-depth knowledge. Subsequently, they collaboratively discussed and answered the given questions. The discussion as shown in Figure1, Figure2 and Figure3.

Question:

1. Construct a Context free grammar G which accepts $N(M)$, where $M = (\{q_0, q_1\}, \{a, b\}, \{z_0, z\}, \delta, q_0, z_0, \phi)$ and where δ is given by

$$\begin{aligned}\delta(q_0, b, z_0) &= \{(q_0, zz_0)\} \\ \delta(q_0, \epsilon, z_0) &= \{(q_0, \epsilon)\} \\ \delta(q_0, b, z) &= \{(q_0, zz)\} \\ \delta(q_0, a, z) &= \{(q_1, z)\} \\ \delta(q_1, b, z) &= \{(q_1, \epsilon)\} \\ \delta(q_1, a, z_0) &= \{(q_0, z_0)\}.\end{aligned}$$

2. Convert the following PDA to CFG

$M = (\{q_0, q_1\}, \{0, 1\}, \{X, Z_0\}, \delta, q_0, Z_0, \Phi)$ and δ is given by

$$\delta(q_0, 0, Z_0) = \{(q_0, XZ_0)\}, \delta(q_1, 1, X) = \{(q_1, \epsilon)\},$$

$$\delta(q_0, 0, X) = \{(q_0, XX)\}, \delta(q_1, \epsilon, X) = \{(q_1, \epsilon)\},$$

$$\delta(q_0, 1, X) = \{(q_1, \epsilon)\}, \delta(q_1, \epsilon, Z_0) = \{(q_1, \epsilon)\}.$$

3. PDA $P = (\{q_0, q_1, q_2, q_3\}, \{0, 1\}, \{X, Y, Z\}, \delta, Z, \{q_3\})$ has the following rules definition δ :

$$\delta(q_0, \epsilon, Z) = \{(q_1, XZ)\} \quad \delta(q_2, 0, Y) = \{(q_2, \epsilon)\}$$

$$\delta(q_1, 0, X) = \{(q_1, YX)\} \quad \delta(q_2, \epsilon, X) = \{(q_2, \epsilon)\}$$

$$\delta(q_1, 0, Y) = \{(q_1, YY)\} \quad \delta(q_1, \epsilon, Z) = \{(q_3, Z)\}$$

$$\delta(q_1, 1, Y) = \{(q_2, Y)\} \quad \delta(q_2, \epsilon, Z) = \{(q_3, Z)\}$$

$$\delta(q_2, 1, Y) = \{(q_2, Y)\}$$

4. Convert the PDA $P = (\{p, q\}, \{0, 1\}, \{X, Z_0\}, \delta, q, Z_0)$ to a CFG. If δ is given by:

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$$1. \delta(q, 1, Z_0) = \{(q, XZ_0)\}.$$

$$2. \delta(q, 1, X) = \{(q, XX)\}.$$

$$3. \delta(q, 0, X) = \{(p, X)\}.$$

$$4. \delta(q, \epsilon, X) = \{(q, \epsilon)\}.$$

$$5. \delta(p, 1, X) = \{(p, \epsilon)\}.$$

$$6. \delta(p, 0, Z_0) = \{(q, Z_0)\}.$$

5. How PDA is converted into CFG? Convert the following PDA into CFG.

$P = (\{p, q\}, \{0, 1\}, \{Z, X\}, \delta, p, Z, \Phi)$

$$\delta(p, 1, Z) = \{(p, XZ)\}, \delta(p, \epsilon, Z) = \{(p, \epsilon)\} \quad \delta(p, 1, X) = \{(p, XX)\},$$

$$\delta(q, 1, X) = \{(q, \epsilon)\}, \delta(p, 0, X) = \{(q, X)\}, \delta(q, 0, Z) = \{(p, Z)\}$$

6. Construct a Context free grammar G which accepts $L(M)$, where

$M = (\{q_0, q_1\}, \{a, b\}, \{z_0, z\}, \delta, q_0, z_0, \Phi)$ and where δ is given by

$$\delta(q_0, a, z_0) = \{(q_0, zz_0)\}$$

$$\delta(q_0, a, z) = \{(q_0, zz)\}$$

$$\delta(q_0, b, z) = \{(q_0, \epsilon)\}$$

$$\delta(q_1, b, z) = \{(q_1, \epsilon)\}$$

$$\delta(q_1, \epsilon, z) = \{(q_1, \epsilon)\}$$

$$\delta(q_1, \epsilon, z_0) = \{(q_1, \epsilon)\}.$$

• Images / Screenshot of the practice:



Figure 1: Students discussion with their team members

$$\begin{aligned}
 3. \quad V &= (S_1 [q_0, a, q_0], [q_0, a, q_1], \\
 &\quad [q_1, a, q_0], [q_1, a, q_1], \\
 &\quad [q_0, z_0, q_0], [q_0, z_0, q_1], \\
 &\quad [q_1, z_0, q_0], [q_1, z_0, q_1]) \\
 S &\rightarrow [q_0, z, q_0] \\
 &\quad [q_0, z, q_1] \\
 \delta(q_0, a, z_0) &= (q_0, a, z_0) \\
 [q_0, z_0, q_0] &\rightarrow a [q_0, a, q_0] [q_0, z_0, q_0] \\
 [q_0, z_0, q_0] &\rightarrow a [q_0, a, q_0] [q_1, z_0, q_0] \\
 [q_0, z_0, q_1] &\rightarrow a [q_0, a, q_0] [q_0, z_0, q_1] \\
 [q_0, z_0, q_1] &\rightarrow a [q_0, a, q_1] [q_1, z_0, q_1] \\
 \delta(q_0, a, a) &= (q_0, a, a) \\
 [q_0, a, q_0] &\rightarrow a [q_0, a, q_0] [q_0, a, q_0] \\
 [q_0, a, q_0] &\rightarrow a [q_0, a, q_1] [q_1, a, q_0] \\
 [q_0, a, q_1] &\rightarrow a [q_0, a, q_0] [q_0, a, q_1] \\
 [q_0, a, q_1] &\rightarrow a [q_0, a, q_1] [q_1, a, q_1] \\
 \delta(q_0, b, a) &= (q_1, a) \\
 [q_0, a, q_0] &\rightarrow b [q_1, a, q_0] \\
 [q_0, a, q_1] &\rightarrow b [q_1, a, q_1] \\
 \delta(q_1, b, a) &= (q_1, a)
 \end{aligned}$$

$$\begin{aligned}
 [q_1, a, q_0] &\rightarrow b [q_1, a, q_0] \\
 [q_1, a, q_1] &\rightarrow b [q_1, a, q_1] \\
 \delta(q_1, a, a) &= \delta(q_1, \epsilon) \\
 [q_1, a, q_1] &\rightarrow a \\
 \delta(q_1, \epsilon, z_0) &= (q_1, \epsilon) \\
 [q_1, z_0, q_1] &\rightarrow \epsilon
 \end{aligned}$$

Figure 2: Answers given by Poojasree and her team members

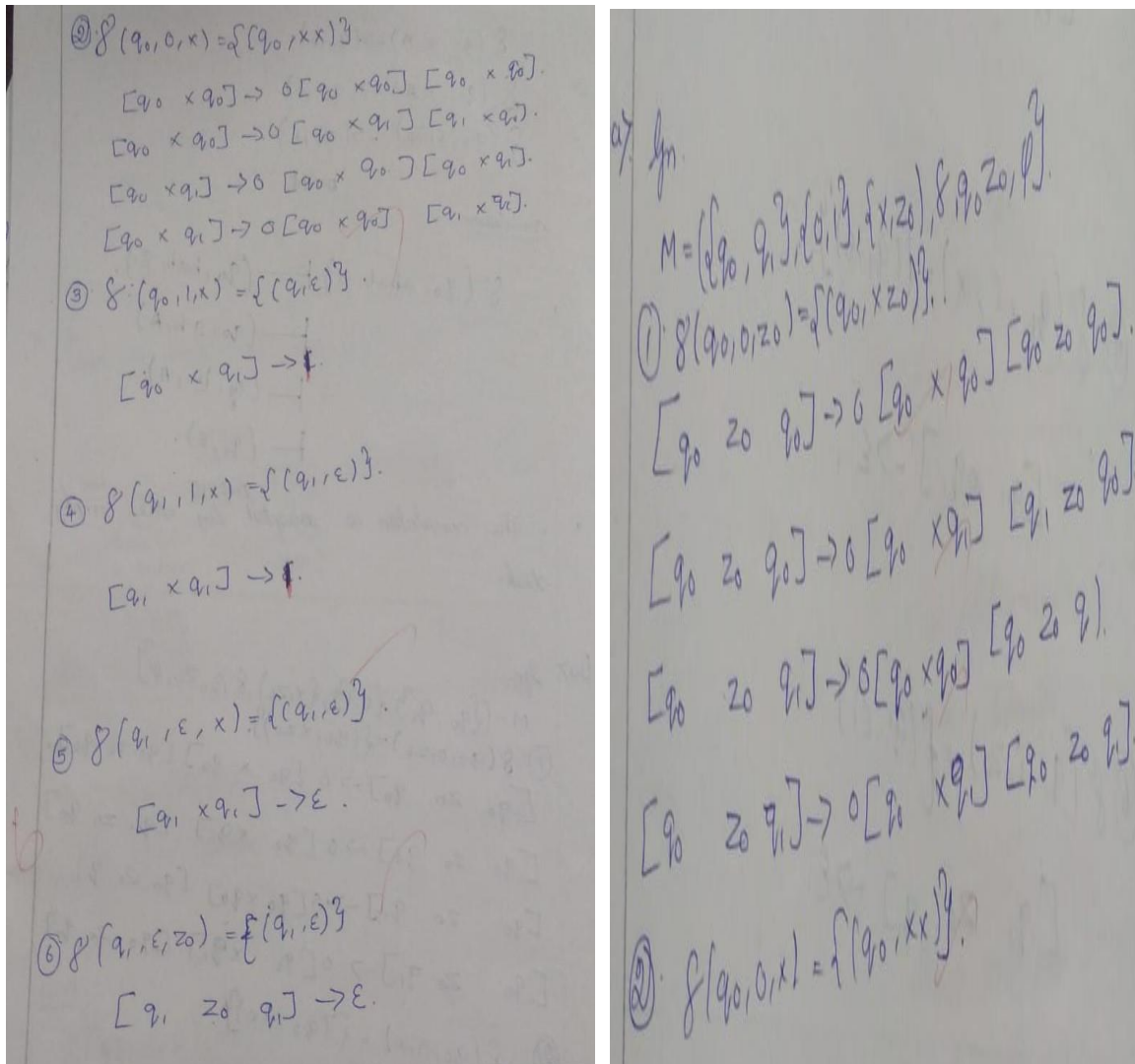


Figure 3: Answers given by Sudharsan S M and his team members

• **CO – PO / PSO mapping: PO / PSO mapped:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO3	3	3	3	1	2	1	2	1	2	2	2

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
	3	3	3	1	2	1	2	1	2	2	2
Justification for correlation	Apply basic Knowledge of mathematics for converting PDA to CFG	Identify the need for converting PDA to CFG	Able to design solutions for complex computational problem by implementing PDA to CFG.	Able to analyze the relationship between the PDA and CFG	Able to prove the PDA and CFG are equivalent	Able to find the solution individually	Based on the given problem, the answer can communicate to a group of peoples	Ability to recognize the need for converting PDA to CFG in real world applications	Exhibit knowledge in PDA to CFG solution for complex computational problems	Exhibit knowledge in PDA to CFG in the field Data Analytics to provide IT solutions.	Exhibit knowledge PDA to CFG in software projects to deliver a quality product

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- The activity helped students better understand the step-by-step conversion of PDA to CFG through peer explanation and discussion.
- Students reported that learning from peers made complex concepts easier to grasp compared to traditional lecture methods.
- Many students appreciated the opportunity to become “experts” in a specific subtopic, which improved their confidence and sense of responsibility.
- Faculty observed increased interaction, collaboration, and knowledge sharing among students.

- ***Benefit of the practice:***

- The students actively participated in this activity.
- Through this activity, Collaboration and teamwork skills were strengthened through group discussions and knowledge sharing.
- It helped in reducing fear and hesitation in handling complex theoretical concepts.
- Students developed a clear understanding of the step-by-step conversion from PDA to CFG through peer teaching and discussion.
- Overall, the activity made learning more engaging, interactive, and student-centered.

- ❖ ***Challenges faced in implementation:***

- Some students lacked prerequisite understanding of PDA or CFG, which made it difficult for them to actively contribute in expert discussions.
- The allotted time was sometimes insufficient for in-depth discussion of all subtopics and for completing the full conversion process

References:

- ❖ <https://www.jigsaw.org/>
- ❖ <https://www.readingrockets.org/classroom/classroom-strategies/jigsaw>
- ❖ https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/VA_CS3352_Jigsaw.pdf

Signature of Faculty Member

HOD